

Basic Commands of Linux

- Commands intro

Kya tamaam linux commands languages C mein bani hain?

Nahi, Linux ki tamam commands sirf C language mein nahi banti, lekin: Most core commands and tools (like ls, cp, mv, ps, cat, grep, etc.) — jo Linux ke core utilities hain — C language mein likhe gaye hote hain. Ye commands GNU Core utils ka part hain, aur C ek fast, low-level language hone ke nate OS-level programming ke liye best mani jaati hai.

Detail Explanation:

C language ka use: Linux ka kernel bhi C mein likha gaya hai.

Core commands like:

ls, cp, rm, mkdir, chmod, chown, cat, echo, etc.

Ye sab C language mein likhe gaye hain.

Source code: <https://github.com/coreutils/coreutils>

Lekin har command C mein nahi hoti:

Bahut si Linux commands aur tools:

Python mein (e.g., yum, dnf, anaconda installer)

Bash script mein (e.g., /etc/init.d/ ke scripts)

Perl, Go, Rust, ya C++ mein bhi likhe gaye hain.

Examples:

Command/Tool	Language
ls, cat, rm	C (coreutils)
yum, dnf	Python
htop	C
nmap	C++
firewalld	Python
docker	Go
ripgrep (rg)	Rust

To sum up:

Most system-level commands in Linux are written in C, but Linux allows tools in any language — Python, Go, Bash, etc.

ls = ka matlab hota hai: list files/directories

- yeh command list k liay use karty hai

ls -l

- Yeh command current directory mein jo bhi files aur folders hain, unki detailed information dikhata hai.

ls -color

- yeh folder ko show karta hai

ls -color

- yeh file ko show karta hai

df = disk free

- yeh command partition ko show karty hai

df -h

- Yeh command disk usage ko human-readable format (GB/MB) mein dikhata hai.

date

- ye command date ko show karty hai

pwd = print working directory

- ye command show karty hai k me current iss path me hu

Example

/muddassir/ali/pak

cd = change directory

- ls command ka use terminal mein ek directory (folder) se doosri directory mein jaane ke liye use hota hai.

mkdir = make directory

- is command se hum folder bnaty hai

Example

mkdir folder name

touch

- is command se hum file bnaty hai

Example

new touch file se file ka size 0

touch file name

rm = remove

- Ye command files ya directories (folders) ko delete karne ke liye use hoti hai
- Example

rm file ya folder name

rm -rf

- Force delete without confirmation

ctrl+c = "Interrupt" ya "Cancel current process" in the terminal

- ye command next jane k kiay istmal hoti hai

whatis

- ye kise command ki short history k liay istmal hoti hai
- Example

whatis command name

man

- ye kise command ki long history k liay istmal hoti hai
- Example

man command name

--help

- ye kasi be command ki flag histroy k liye istmal hoti hai
- Example

command name --help

Question for kanowledge

1.
 - Linux kyu best hai window se?
2.
 - Linux me virus kyu ni ata ?
3.
 - Linux me zyada terminal kyu hote hai?

First Lecture 01

```
touch file name {1..20}
```

- is se hum jitne chaye file bna skte hai

Linux file system Hierachy consepts (FSHC)

```
/
```

- root ya main diercoty hote hai

Absolute path

```
File ya folder ka full location starting from / (root directory)
```

- Example

```
/home/umar/documents/file.txt
```

relative path

```
File ya folder ka rasta jo aapki current location (current directory) ke hisaab se diya jata hai.
```

Yeh absolute path ke opposite hota hai.

- Example

```
/home/umar/
```

Recap of Commands

- ls ,pwd, touch, mkdir, rm , whatis , man , --help

```
su - username
```

- is comman ki madat se hum user change karte hai

```
ls -d
```

- Folder ka sirf naam dikhana, uske andar kya hai wo nahi dikhana
- or yeh be list check karta hai

```
mkdir -p
```

- iss ko use kar k hum direct koi be folder k ander k folder create kar skte hai jo phaly na create hu .jesa k ali/2025 folder na ho or pak iss ma bnana hu

Example

```
mkdir -p /opt/ali/2025/pak/
```

Revision commands / flags continued

- Linux is multiuser mode & windows is not multiuser mode.

- we cannot use simultaneous both users in windows but linux we can work on multiple terminals.

id or whoami

- is se hum find karte ha k humary pass current user kon hai

cat /etc/passwd

- ye new user ka passwd show karta hai or iss se hum apne user dikh skty hai

cat = cat ki madat se hum kasi be file ko read karskty hai

useradd

- is command ki madat se hum new user create karte hai

passwd username

- is command se hum user ka new passwd select karte hai

which command

- is se hum check karte hai yeh command kon se path per hai

__ Example __

which command name

find / -name

- is command se exact file ka name likh kar find kar skty hai

__ Example __

- find / -name "file name"

Overview redheat linux directories

/ |— /bin → Basic commands (ls, cp, mv, rm)
 |— /boot → System boot files (kernel, grub)
 |— /dev → Devices (hard disk, USB, printer)
 |— /etc → Configuration files (system settings)
 |— /home → User data (har user ka apna folder)
 |— /lib → Libraries (jaise Windows ke DLL files)
 |— /lost+found → Crash ke baad recovered files
 |— /mnt → Temporary mount (USB, CD, extra drive)
 |— /media → Auto mount (USB/CD insert karne par)
 |— /opt → Extra software (3rd-party apps)
 |— /proc → System & process info (virtual files)
 |— /root → Root user ka home folder
 |— /sbin → System admin commands (shutdown, mount)
 |— /srv → Service data (web, FTP ke liye)
 |— /tmp → Temporary files (auto delete on reboot)

└─ /usr → User programs, software (applications)

└─ /var → Variable files (logs, cache, mail)

Easy Memory Trick (Short Notes):

/bin → Basic commands

/boot → Boot files

/dev → Devices

/etc → Settings

/home → Users' data

/lib → Libraries

/mnt → Mount drives

/opt → Optional software

/tmp → Temporary files

/var → Logs & cache

Editors

note

Linux me bedefault file txt bante hai lekn show ni hoti

- - vim ka istmal

```
vim :set nu
```

- ye file me likhe data ki line ka btata ktne hai

```
:set nonu
```

- ye file me likhe data ki line ko khatam karta hai

```
/(jo word find karna hai wo enter karen)
```

- split window me koi word find karna hu to use karte hai
 - n next word per jane k liye use huta hai
 - shift+n back word per ane k liye

```
ctrl+ww
```

- ▪ ye window jane k liye

```
:split /etc/hosts
```

- iss ki madat se ek or file open karskte hai

`:vsplit`

- ye data k do part karta hai
- - `nano filename`
 - ye be file create karta hai

`vim`

- ye file ma data likhen k liye istmal huta hai

`:wq`

- is se file ko save kia jata hai, save and exit

`shift+X`

- is se hum file me passwd select karte hai but file vim me open honi chayee

`:set key = or : enter enter`

- ye command passwd ko remove karta hai

`yy`

- ye command line ko copy karne k liye istmal hute hai

`p`

- ye paste k liye istmal huta hai

`dd`

- ye command line ko delete karne k liye istmal hute hai

`x`

- ye read do k liye istmal huta hai for word

`u`

- ye undo k liye istmal huta hai

`:%s /name / replace name/g`

- ye vim or vi editor me word ya name ko replace karne k liye istmal huta hai or word "g" se all replace hujata hai puri file me se

`vi file name`

- file ko opne kia jata hai

`:X`

- yeh vim me file ko passwd dene k liye istmal huta hai

task

vim me passwd to set kia hai mager remove kasye karen

2 enter press karen

Lecture no 7

- > Soft link [symboli] and hard link

- cp and mv commands

In

- ek Linux/Unix command hai jo hard link ya symbolic link (symlink) banane ke liye use hoti hai.

In -s [filename] [link file name] -> asal file ka pta

- ye command soft link bnate hai .is ka matlab hai ek or file

In [filename] [link file name] -> asal file dosra naam

- ye command hard link bnate hai hard link me ager parent file delete hu to child me data rhta hai

ls -l childfile name

- ye check karta ha k child file bane hai k ni

delete file ko "orphan " khte hai or iss file ka color red hujata hai

cp [file name] [paste path]

- ye command file ko copy karne k liye istmal hute hai

cp -r [folder name] [paste path]

- yeh command -r recursive hai jo k puri folder copy paste karne k liye use huta hai

mv [file name] [paste path]

- ye command file or directory ko move karne k liye istmal huta hai or ye cut and paste ka kam be karta hai

mv [old file name] [rename file]

- ye file ka naam be rename karta hai

Question for interview

- different b/t soft link and hard link

advanatges soft link

- easy to access
- soft link can cross partition and can also be created over dirs.

disadvanatges

- if parent get delete child become orphan - no use

advanatges of hard link

- type of backup
- if parent is deleted , no impact on child file

disadvanatges

- cannot cross partition cannot be created over dirs.

Inode

- every file will have a unique inode number .
- inode keep attributes of a file and properites
- her file ka ek unique inode number huta hai or all data file k inode me save hota hai.file name only pachan k liye huta hai

ls -il

- ye file k inode number k sath puri details show karta hai
 - hard link file inode number will be same
 - soft link file inode number will be different
- Iska matlab hai ke file.txt ka data inode #12345 ke andar stored hai. Agar aap "ls -i" command use karte hain, to aap file ka inode number dekh sakte hain.

stat filename

- ye inode k sath all details show karta hai

_ - stat ki details interview ma askty hai

- assess = atime
- modify time = mtime
- changetime = ctime

how to create hidden file

touch .filename

- dot file ko hidden karne k liye use huta hai

how to see hidden file

```
ls -a
```

how to see hidden file inode number

```
ls -la
```

how to see file inodes

```
ls -li filename
```

version check

- cat /etc/redhat-release
- cat /etc/os-release
 - yeh cloud me version check karen k liye istmal huta hai

```
arch
```

- iss ki madat se hum architect check karte ahi

```
example : X86_64
```

```
uname --help
```

- is command ka flag check kar k uname se device ki information dekhe hai
 - example: version,kernal,architect etc.

Ownership and permission

- ownership and permission me humary pass 9 field hute hai

```
-rw-r--r-- . 1 root root 0 may 10 07:50 /opt/muddassir
```

☐ ye type of file

2 - [rw-r--r--] ye permission hai

```
rw-r--r--
```

- iss me 9 bit hai, --- bit off , rwx, bit on -w-first off second on thrid off

3 - [.] ye selinux hai

4 - [1] link count = ye hard link me change huta hai 2 be hu skta hai or 3..

5 - [root] owner or user

6 - [root] ye group hai

- **tip**
 - jb hum ek user create kaarty hai us k sath same naam ka group auto create hujata hai

7 - [0] size of bytes mean size of data

8 - [may 10 07:50] accesstime

9 - [/opt/muddassir] file name

Permission ki qismein:

Read (r): File ko parhne ki ijazat hoti hai.

Write (w): File me tabdeeli karne ya directory me files add / delete karne ki ijazat hoti hai.

Execute (x): File ko run karne ya directory ke andar enter hone ki ijazat hoti hai.

Permission ki satahein

User (u): File ka malik (owner).

Group (g): File ka group jiske users ko access milta hai.

Others (o): System ke baqi tamam users.

```
ls -ld [file name]
```

- ye permission k liye command istmal hute hai

```
--- bit off ko show karta hai
```

```
rx bit on ko show karta hai
```

Binary permission

- ■ 000 permission off
- ■ 111 permission on

how to change onwership

```
chmod [ kiss ki permission change karni hai user,group and other ] [ file name]
```

how to change owner

```
chown [name of user] [ file name]
```

how to change groups

```
chgrp [ name of group ] [ file name]
```

how to add group

```
groupadd [ group name]
```

- ■ user only sigle
- ■ group have multipul user

rwX	rwX	rwX
u	g	o
user	group	other

how to change permission of user

symbolic way

```
chmod u+rw file name
```

- ◦

```
chmod u-rwx filename
```
- ◦

```
chmod g+rw filename
```
- ◦

```
chmod ugo+rw filename
```

octal way of permission

- user

read	write	execute
4	2	1

- ◦ ye user ,group and other sub ki 7 7 7 permission hai
- group

read	write	execute
4	2	1

- other

read	write	execute
4	2	1

7 types of - file of type in permission [" - " ye type of file hai iss k 7 types hai]

- [-] ager ata ha to regular file hai
- [d] ager ata ha to directory

- [l] ager ata ha to symlink
- [b] ager ata ha block file
- [c] ager ata ha chareter file
- [s] ager ata ha socket file
- [p] pipe device file

Size in bytes

1 Byte (B) = 8 bits

1 KB (Kilobyte) = 1024 Bytes

1 MB (Megabyte) = 1024 KB

1 GB (Gigabyte) = 1024 MB

1 TB (Terabyte) = 1024 GB

1 PB (Petabyte) = 1024 TB

1 EB (Exabyte) = 1024 PB

1 ZB (Zettabyte) = 1024 EB

1 YB (Yottabyte) = 1024 ZB

Advance permission Lecture

Inherit Permission

chattr = change attribute

+i = immutable

```
chattr +i filename
```

- ye file ko immutable karta hai "a" ye append karta hai
 - ye use karne k bad file delete ni hoskte

```
chattr -i filename
```

- doobara phaly halet me karta hai

```
lsattr filename
```

- is se check karte hai k chattr file create huge hai k nahi

1- file should have only read permission

2 - group should only have execute permission

3 - other should not have any permission

```
setfacl -m u: username :rwx file
```

- is command se hum her ek user ko alg alg permission deskte hai

```
getfacl filename
```

- is se pta chalta ha k permission set huge hai

```
setfacl -x u: username :rwx file
```

- remove karne k liye

```
chmod -Rv 777 (folder name)
```

- recursive ki madat se puri folder ki permission change kar skte hai. puri folder me kitne bhi file huge sub ki permission change hugay ge

internet tool for chmod

- internet per ek tool hai "chmod calculator" k name se

end permission topic

runlevel 0 to 6 =total 7

windows mode

- start normally mode
- windows with safemode

- hum hamesha normally mode me kam karte hai
- iss ko hum runlevel khte hai aur 7/8 me target khte hai
- ```
runlevel
```
- ye check karta hai current kis phalor per hai

**Runlevel** : me 6 konsol hai or humari pass total 6 konsol hote hai

multi user me # runlevel 3

- is me humari pass 6 terminal hote hai jo hum "alt+f1,f2,f3,f4,f5,f6" se open kar sakte hai

- ○ init command

- ○ is ki madat se runlevel switch karte hai
- init ye ek lift ki tarah kam karta hai

## Runlevel kya hai?

- Runlevel ek predefined mode hai jo batata hai ki Linux system boot hone ke baad kis stage me operate karega — jaise ki graphical mode, multi-user mode, single-user mode, etc.
- ○ Runlevel basically system ke startup behavior ko control karta hai.

### Common Runlevels (SysV init systems):

```
Runlevel Description
>0 Halt / Shutdown

>1 Single-user mode (Maintenance mode)

>2 Multi-user mode (No network - rarely used)

>3 Full multi-user mode (Command-line only) (vcs virtual consoles)

>4 Not used/User-defined

>5 Multi-user mode with GUI (Graphical Mode) gnome terminal
. gnome terminal pts huta hai pts (sudo terminal)

>6 Reboot
```

### Runlevel 3 me 6 terminal hute hai jin ko vcs (vitural consoles system). tty

- ○ alt+f1 ,alt+f2 ,alt+f3, to alt+f6 TTY (teletypewriter devices) tty terminal hute hai

#### ps command

- ye process stauts show karta hai
- or yeh command show karte hai k app kiss tty1 ya tty2 me hai

### Runlevel 5 me 400 terminal hute hai jin ko gnome terminal khte hai. pts

- yeh gnome terminal hute hai gui (graphical user interface) pts (psudo terminal ) pts yeh runlevel 5 me huta hai

#### ls /dev/pts or ls /dev/tty

- iss ki madat se hum status check karte hai k humary pass ktne tty ya pts hai
  - tip

- putty also use pts but putty graphical load ni karta ( pts stand for psudo terminal ) yeh genome terminal me huta hai

## Note (Systemd-based Systems like Ubuntu 16+, CentOS 7+8, etc.)

- Modern Linux distros Systemd use karti hain, jisme runlevels ki jagah targets use hote hain.

| Traditional Runlevel | systemd Target    |
|----------------------|-------------------|
| 0                    | poweroff.target   |
| 1                    | rescue.target     |
| 3                    | multi-user.target |
| 5                    | graphical.target  |
| 6                    | reboot.target     |

- ◦ ager hum runlevel 3 me hai to temprary GUI open karskte hai

```
start x
```

- ye hum ko temprary GUOI open kar deta hai

```
pkill x
```

- ye hum ko dobara terminal 3 me open kar deta hai

```
cat /etc/inittab
```

- ◦ is me by default show huta hai k humara system kon se runlevel me jay ga reboot hune k bad.

```
shutdown -r now
```

- reboot karta hai

```
vi /etc/sysconfig/init
```

- iss ki madat se hum centos 6 me runleve 1 ko bolock kar skte hai

## Targets

### Change target (permanently):

```
systemctl set-default graphical.target -= 5
```

```
systemctl set-default multi-user.target -= 3
```

- ```
systemctl get-default
```

 - iss ki madat se hum check karte hai k default. gui hai k cli hai

Targets in hrel 7/8

iss me major 2 targets hute hai

```
systemctl isolate graphical-user.target
```

yeh use hote hai 7/8 me

```
systemctl isolate multi-user.target
```

all sep recover passwd in rhel 7 & 8

```
# RHEL 7/8 passwd reset while booting

go to splash screen -interrupt
press e
linux 16
rd.break
ctrl x
mount -o remount rw/sysroot
chroot /sysroot
passwd root
touch /.autorelabel
exit
exit
```

task question

- hard disk me kum space que show hote hai?

ek cylinder ka size = 8MB hai

example

$1958 * 8 \text{mb} / 1024 = 15 \text{GB}$

bc =ye calulator in linux

11 lecture

fdisk kya hai?

- fdisk ek command-line tool hai Linux/Unix systems me jo hard disk ki partitions banane, dekhne, modify karne ke liye use hota hai.
- Yeh mainly MBR (Master Boot Record) based disks ke liye kaam karta hai.

hard disk k part hute hai

```
[root@server1 Desktop]#
[root@server1 Desktop]# vi rootpassword_reset_steps_RHEL7_8.txt
[root@server1 Desktop]#
[root@server1 Desktop]#
[root@server1 Desktop]# cat rootpassword_reset_steps_RHEL7_8.txt
```

RHEL 7 /8 passwd reset while booting

go to splash screen - interupt

press e

```
linux16 .....
..... rd.break
```

ctrl x

```
SH# mount -o remount rw /sysroot
```

```
SH# chroot /sysroot
```

```
SH# passwd root
```

```
-
-
```

```
SH# touch /.autorelabel
```

```
exit
```

```
exit
```

uper wala = magnetic us k bad wala =track us k bad wala = sector

device driver kya hai

- ■ device driver low language program hute hai
- ■ ■ with interact b/t hardware and respective software
- jo hum ne partaition banai hai ye sdb us k sath run hute hai

every partation is associate with driver

- jb koi be partation bnate hai woh ak driver k sath attach hujate hai
- ■ example sda,sda1,sdb,sdb1 etc.

fdisk kis kaam me aata hai?

- ◦ New partition create karne ke liye
- ◦ Existing partition ko delete ya modify karne ke liye
- ◦ Disk layout dekhne ke liye
- ◦ Bootable flag set/unset karne ke liye

__ fdisk ka basic use: __

```
fdisk /dev/sdX
```

- Yahaan /dev/sdX me:

X = a, b, c... (jaise /dev/sda, /dev/sdb)

```
fdisk /dev/sda
```

Example: New Partition Banana

```
fdisk /dev/sdb
```

Then:

Press n → new partition

Press p → primary

Choose partition number (1-4)

Set first and last sector (ya default enter karein)

Press w → save and exit

for useable partition next steps:

```
partx -a /dev/sdb > kernel ko refresh karen  
mkfs.ext4 /dev/sdb1  
mkdir directoryname  
mount /dev/sdb1 /directoryname
```

df -Th = ye check karta partition ko dekhne k liye

why mount ?

- data ko access karne k liye partition se

6 -me partx -a or 7 - me partprob -s huta hai

karnel table kya hai or karnel table ram me huta hai

```
partx -a /dev/sdb
```

- ye command ko use karne se karnel table ko update karta hai jb karnel table update huta hai to os me entry hu jate hai jis ki madat se hum dekh skte hai partition bange hai
- - Ager hum ne partition table ko update ni kya hai lekn karnel table ko update karne se phaly system reboot huga hai to hum ko karnel table ko update karne ki zarorat ni hugi or wo os me

update hojay ga

```
fdisk -l /dev/partition ka name jese (sdb,sda,sdc ect.)
```

yeh command use karne se jo disk banai hai wo show karta hai

Partition table kya huta hai

- jo hum partition bnane k bad save karte hai "w" se to woh partition table ko update karta hai

```
fdisk -l
```

- ye show karta hai humary pass ktne hard disk hai

```
ls /dev/sd*
```

- check harddisk diver

Hard Disk kya hai Linux me?

- Linux (ya kisi bhi OS) me Hard Disk ek physical storage device hoti hai jisme data (OS, files, programs) store hota hai.
- - Linux isse ek device file ke roop me dikhata hai, jaise:

```
/dev/sda - pehla hard disk
```

```
/dev/sdb - doosra hard disk
```

```
/dev/nvme0n1 - NVMe SSD drives ke liye
```

```
#echo "----" > /sys/class/scsi_host/host0/scan
```

```
#echo "----" > /sys/class/scsi_host/host1/scan
```

```
#echo "----" > /sys/class/scsi_host/host2/scan
```

- - yeh comman iss waqat use karte hai jb system on hu or hum ne partition create ki hai or show ni hurhe or system ko reboot be ni kar skte us waqat use hute hai

Partition kya hota hai?

- Hard disk ko chhote-chhote logical parts me divide kiya jata hai, jise partition kehte hain.
- - Har partition ko Linux me alag device ke roop me dikhaya jata hai:

```
/dev/sda1 - pehla partition of /dev/sda
```

```
/dev/sda2 - dusra partition
```

```
/dev/sdb1 - pehla partition of dusri disk
```

Kitne types ke Partition hote hain?

- Partitioning system pe depend karta hai:
- fdisk → MBR disk ke liye
- parted → GPT + large disks (recommended)
- cfdisk → menu based (easy)
- **1. MBR (Master Boot Record) – purana system**
 - ◦ Max 4 primary partitions
 - ◦ Ya: 3 primary + 1 extended (jisme multiple logical partitions ho sakti hain)
- **2. GPT (GUID Partition Table) – naya aur modern**
 - ◦ 128 partitions tak support karta hai (Linux me)
 - ◦ Har partition primary hota hai (no extended/logical needed)
 - ◦ Large disks (>2 TB) ke liye zaroori hai

Types of Partitions (Usage-wise)

- Partition Type Use Case

```
/ (root)    Main Linux system files
/home      User data
/boot      Boot files (kernel, initrd)
swap       Virtual memory
/var       Log files, mail, web server data
/tmp       Temporary files
/opt       Optional software
```

file system or inode number (index number)

Aasan misaal:

Socho ek library hai (Hard Disk):

Library me kitabein rakhi hoti hain (Files)

Ek register hota hai jisme likha hota hai: kitab number, musannif (author), shelf number

- ye Index (Inode) hota hai

Asal kitabein shelves par rakhi hoti hain

- ye Data Blocks hote hain

Library ka rule book hota hai jisme bataya hota hai:

kitni shelves hain

kitni khaali hain

- ye Superblock hota hai

Khulasa (Summary):

File System = wo structure jo disk par data ko manage karta hai

Includes = Boot Block, Superblock, Inode Table, Data Blocks, Directory Structure

Index = file ya directory ke liye Inode number / record jo asal data ka rasta (location) batata hai

Question:

- different b/w file system

ext2,ext3,ext4 and xfs

ext2, 4 version ext3, 5 version ext4 , 6 version xfs , 7/8 version

different b/w file system

Common File Systems (FS types)

Linux: ext2, ext3, ext4, XFS, Btrfs

Windows: FAT32, exFAT, NTFS

macOS: HFS+, APFS

File System	Journal (iss se murad h reboot k bad recove na huna)	Max FS Size	Max File Size	Use Case
ext1	No	2 GB	Few MB	Very old, ab use nahi hota
ext2	No	32 TB	2 TB	USB/Flash storage
ext3	Yes	32 TB	2 TB	Stable, old Linux systems
ext4	Yes	1 EB	16 TB	Default, fast, general purpose
XFS	Yes	8 EB	8 EB	Large files, enterprise, CentOS/RHEL

Summary

ext2 = simple aur journal-less (USB, flash ke liye best).

ext3 = ext2 + journal (stable).

ext4 = modern, fast, large file support (default in most Linux).

XFS = enterprise level, big files ke liye best (RHEL/CentOS default).

> vi /etc/fstab

file system table me field

interview Question

```
cat /etc/fstab
```

- fstab is a boot process file

/etc/fstab File ke andar kitni fields hoti hain?

Total: 6 fields (columns)

- Har line (entry) me 6 fields hoti hain, separated by spaces ya tabs.

Fields of /etc/fstab explained:

Field	Field Name	Description
1	Device	Device name ya UUID (e.g. /dev/sda1, UUID=xxx)
2	Mount Point	Jahan partition mount hoga (e.g. /, /home, /mnt/data)
3	Filesystem Type	Filesystem ka type (e.g. ext4, xfs, vfat, swap)
4	Mount Options	Mount karte waqt use hone wale options (e.g. defaults, noatime, ro, rw)
5	Dump	Backup ke liye use hota hai (0 = disable, 1 = enable)
6	Fsck Order	Boot time pe filesystem check order (0 = no check, 1 = first check, 2 = later)

Example entry of /etc/fstab:

```
UUID=1a2b3c4d /          ext4    defaults      1 1
UUID=5e6f7g8h /home      ext4    defaults      0 2
/dev/sda3      swap        swap    defaults      0 0
```

Explanation of Example:

- / partition ext4 filesystem hai, defaults options ke saath mount hoti hai, aur fsck order me pehle check hoti hai.
- /home dusri partition hai, wo second fsck order me check hoti hai.
- swap ko fsck aur dump ki zarurat nahi hoti, isliye dono 0.

Summary:

Field	Example	Meaning
1	/dev/sda1	Partition ya device
2	/	Mount point
3	ext4	Filesystem type
4	defaults	Mount options
5	1	Dump backup (1 = yes)
6	1	fsck check order (1 = first)

```
fstab
```

- ye file system table hai

Check current disk & partitions:

```
lsblk
```

- list block devices:

```
parted -l
```

- – GPT & MBR dono detect karta hai:

Example Output of lsblk:

```
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda          8:0    0 100G  0 disk
├─sda1       8:1    0   50G  0 part /
├─sda2       8:2    0   45G  0 part /home
└─sda3       8:3    0    5G  0 part [SWAP]
```

Summary:

- Hard disk = physical storage (Linux me /dev/sda, /dev/sdb, etc.)
- Partition = logical part (Linux me /dev/sda1, /dev/sda2, etc.)
- MBR → 4 primary partitions
- GPT → 128 partitions
- Linux me har partition ka specific role ho sakta hai: /, /home, swap, etc.

Hard disk part

-
- sata - ssd - sad - pendrive -san -nas

Hard disk inside parts pic

click-here

Note:

- fdisk GPT disks (GUID Partition Table) ko handle nahi karta. Uske liye aap gdisk ya parted use karte hain.

Task Question:

- how to hard disk works inside hard disk 10 mints animated video
 - how to data is witten i dard disk types
 - san working
 - nas working
-

task

- add disk from vm of 800 mb
- discover hard disk inside your vm
- fdisk --- create partition table

Compression Tools

Compression Tools kya hain Linux me?

Definition:

- Compression tools aise software hote hain jo files ya folders ko compress (chhota) karte hain, taki unka size kam ho jaye aur storage ya transfer aasaan ho jaye.

gzip ,bzip2 and xz

- yeh compression tools jo kise be file ya folder ka size kam karne k liye istmal hute hai

compression technology kya hute hai?

- - iss ko bolte hai shrink karna or technically bolte deflate karna
- - jis se file ya folder ka size kam huta hai



Popular Compression Tools in Linux:

Tool	Extension	Description
------	-----------	-------------

Tool	Extension	Description
gzip	.gz	Fast and common tool for compressing single files
gunzip	(uncompress)	gzip se compressed file ko extract karta hai
bzip2	.bz2	Better compression than gzip but slower
bunzip2	(uncompress)	bzip2 se compressed file ko extract karta hai
xz	.xz	Very high compression ratio
unxz	xz se compressed file ko decompress karta hai	
zip	.zip	Windows compatible, compress multiple files
unzip	zip files ko extract karta hai	
tar	.tar	Archive tool, multiple files ko ek me bundle karta hai
tar + compression	.tar.gz, .tar.bz2, .tar.xz	Archiving + Compression

gzip filename

- ye file ko compress karta hai

gunzip filename

- ye file ko uncompress karta hai

zip filename.zip abc.txt abcd.txt

- basically kya kam karta hai ye (abc.txt or abcd.txt) ko filename.zip me compress kar k dalde ga

file or folder create and delete

touch file {1..10}

- in se hum file create karte hai {} in curly braces ko wildcard khte hai

rm dir folder{1..10}

- ye folder ko delete karen k liye istmal hute

rm -rf file*

- all file ya folder ko remove karen k liye istmal hute

redirection symbol {> <}

(> ye symbol redirection hai)

- iss ka kam hai ek file ya folder ka data dosre ko dena

Exmaple cat man > file

- ye kia kam karen ga cat ka data file me transfer karde ga

Example file < cat

- ye kia kam karta hai cat ka data file me transfir karde ga

(>) ye single sing overwrite karta hai

(>>) double sing append karta hai jis ka matlab hai add

- ◦ Example file.txt > file1
- ye kya karta hai file.txt ka data file1 me karde ga or file1 ka apna data delete hujay ga
- ◦ Example file1 >> file.txt
- ye kia karen ga data remove ni karta or add karde ga

1. tar – Archive Tool (Backup & Compression)

Purpose:

- Multiple files aur directories ko ek single archive me bundle karna.
- Optional: compress bhi kar sakte hain.

Commonly used for:

- Backups
- Software packaging
- File transfer ke liye ek bundle banana

Common tar Commands:

Action	Command
Create tar file	tar -cf archive.tar folder/
Create tar + gzip	tar -czf archive.tar.gz folder/
Extract tar	tar -xf archive.tar
Extract tar.gz	tar -xzf archive.tar.gz
List contents	tar -tf archive.tar

Example:

```
tar -czf backup.tar.gz /home/umar/
```

2. rsync – File Synchronization Tool

Purpose:

- Files ya folders ko copy/sync karna efficiently (locally ya over network)
- Sirf changes copy karta hai, pura file nahi — fast and bandwidth efficient.

Commonly used for:

- Backup (incremental)
- Syncing folders between servers
- Remote copy using SSH

Common rsync Commands:

Action	Command
Local folder sync	<code>rsync -av source/ destination/</code>
Remote sync (SSH)	<code>rsync -av -e ssh /local/path user@remote:/path/</code>
Dry run (test)	<code>rsync -av --dry-run source/ dest/</code>

Example:

```
rsync -avz /home/umar/ user@192.168.1.100:/backup/
```

- Is command se /home/umar/ ka backup remote server pe /backup/ folder me chala jayega.

defalte

- is ka matlab ha compress

inflate

- is ka matlab ha decompress

Summary:

- tar = Archive + compression (mostly for packaging and backups)
- rsync = Efficient syncing and copying (for live backup, server sync)

gzip - bzip2 -

- ye size ko kum kare k liye use hute hai

gunzip - bunzip2

- ye unzip k liye use hute hai

symbols

- operators wild character khte hai

cp .rsync .cpio .zip .gzip . bzip2

- yeh backup tools ha in linux

cp

- ye sarif file ko copy karta hai " rsync " is very power full backup tool and rsync is a remote syne .rsync inreemental backup tool

scp

- ye be powerful backup tools ha

rsync

- ye remote machine port ssh ko istmal karta hai

sync

- is ka matlab hai(RAM) se data hard disk me lyna or hard disk ki speed fast hujate hai

Question

What is the reason for os to come in RAM?

what is file system

- a file system collection of metadata file system k ander kia huta hai (inodes, inodetabe,super block mount info)
- mount ==> ye mont temp ha till reboot tak change after reboot houga mount nikal jata hai

interview Question

Backup types -Interview question

- full backup
- incremental backup
- differential backup
- cumulative

Question

how to use rsync with tar ,zip ?

redirection > or >> greanter then

```
command > filename
```

- Yeh command file ko overwrite karti hai (purani content hata kar nayi likhti hai).

Summary:

Operator Kaam

```
Output ko file me likhta hai (overwrite)
```

```
Output ko file me add karta hai (append)
```

Note:

- - `>` overwrite karta hai file content ko.
- Agar aap append karna chahte hain (purani file me kuch add karna), to use karein `>>`.

interview Question

- apache - lamp - mysql - boot - process - suid - stickybiy

lamp

L = Linux (OS)

A = Apache (Web server)

M = MySQL (Database)

P = PHP / Python / Perl (Backend language)

Topci FSCR

- `fscr ==>` file system check and repair

```
badblok -v /dev/sdb1
```

- ye system me error ko door karta hai

```
e2fsck
```

- ye new command hai ye be error door karte hai

```
e2fsck -yc /dev/sdb1
```

- ye all information show karta hai

1. fsck – File System Consistency Check

Purpose:

- File system ki checking aur repairing ke liye generic command.
- Har file system ke liye fsck automatically appropriate tool use karta hai.

Example:

```
sudo fsck /dev/sda1
```

- Yeh /dev/sda1 partition ko scan karega aur agar koi error mile to fix karega.

Note:

fsck is like a wrapper — internally yeh e2fsck, fsck.xfs, etc. jaise tool ko call karta hai based on file system type.

2. e2fsck – ext2/ext3/ext4 File System Checker

Purpose:

- Specifically ext2, ext3, ext4 file systems ke liye use hota hai.
- fsck se zyada direct control deta hai ext file systems ke upar.

Example:

```
sudo e2fsck /dev/sda1
```

- Agar /dev/sda1 ext4 ya ext3 format ka hai, to yeh command us file system ko scan karke errors ko fix karegi.

Difference: fsck vs e2fsck

Feature	fsck	e2fsck
Type	Generic tool	Specific to ext2/3/4
Auto detect	(detects FS type)	(used manually for ext*)
Internally uses	e2fsck, fsck.xfs, etc.	N/A
Usage	All file systems	Only ext-based file systems

Important Tips:

- File system ko check karne se pehle usko unmount karna chahiye:

```
umount /dev/sda1
```

```
fsck /dev/sda1
```

- Never run fsck on a mounted and in-use partition (especially / root) — it can cause data corruption.

Summary:

- fsck = Generic file system check command (all types)
- e2fsck = ext2/ext3/ext4 ke liye specialized tool
- Unmount before using, otherwise damage ho sakta hai.

Task

e2fsck Se File System Fix Karne Ka Tarika

Step 1: Affected Partition Ko Unmount Karen

- e2fsck ko kabhi bhi mounted partition pe mat chalaiye.

```
umount /dev/sdb1
```

Step 2: e2fsck Command Chalaiye

```
e2fsck -p /dev/sdb1
```

- Ya, agar aap manually har prompt ka jawab dena chahte hain:

Important Notes:

- Unmounted partition par hi e2fsck chalayen.
- Root partition pe agar chalana ho, to system ko Live CD/USB se boot karen.
- Backup lena hamesha safe hota hai — fsck ya e2fsck se data loss bhi ho sakta hai agar damage severe ho.

```
fdisk -l
```

- Yeh commands aapko pata denge ki /dev/sda1, /dev/sdb1, etc. kaunse partitions aapke system me hain.

What is Universally Unique Identifier(uuid)

UUID ka Use:

- Har partition ya file system ka ek unique identifier hota hai.
- System boot time par ya `/etc/fstab` me reliable mounting ke liye use hota hai.
- UUID physical naming (`/dev/sda1`, etc.) se better hai kyunki device name badal sakta hai, par UUID fix rehta hai.

UUID Check Karne Ka Tarika:

- `blkid` command:
- ya `lsblk -f`

```
/dev/sda1: UUID="4e9f6b18-1a34-4c3c-9e93-888a75f4f0a1" TYPE="ext4" /dev/sda2: UUID="22dfcfd-0c8c-4c5f-a1a3-9b67aee973bc" TYPE="swap"
```

UUID Ka Fayda:

UUID se Mount	Device Name se Mount
Stable (never changes)	Change ho sakta hai (like <code>/dev/sdb</code> → <code>/dev/sdc</code>)
Reliable in <code>/etc/fstab</code>	Risk of wrong device mounting
Better in dual/multi-boot	Less preferred in modern systems

1. Process ID kya hai?

Definition:

- Process ek program ka running instance hota hai. Jab aap koi program run karte ho (jaise firefox, ls, vim, python etc.), to system ek process banata hai jisme wo program execute hota hai.

Example:

```
gedit
```

- `ls` command se `gedit` ek process ban jata hai, jiska apna PID (Process ID) hota hai.

Process dekhne ke liye:

```
ps aux
```

-ya

```
top
```

```
pkill [ filename ]
```

- ye be kill karta hai

kill (pid)

- ye kill karne k liye istmal huta hai

pstree

- ye sara process tree me dekhata hai

=> top psid tools ya monitaring tools

- ps
- pstree
- lsof

: () { : | : \$ } ; :

init aur systemd

- init is the father of all process

Question : Interview question

what is buffers?

Simple Explanation: Buffers = "temporary space for data to be written to disk"

what is cached?

Simple Definition: Cached memory is RAM used to store recently accessed file data for faster reuse.

what is free commands?

Important Point:

- free memory ≠ available memory free column ke numbers misleading ho sakte hain kyunki Linux unused memory ko cache ke liye use karta hai.

-
- 1 **find command 15 example**
 - 2 **ps command 15 example**
 - 3 **rsync command 15 example**

what is PId zero

PID 0 ka matlab:

PID 0 = Swap daemon ya scheduler process

- Ye process system boot hone ke waqt sabse pehle create hoti hai (kernel ke andar).

- Iska kaam hota hai:
- CPU scheduling manage karna
- Idle process handle karna (jab koi aur process nahi chal raha hota)

Details:

Feature	Description
PID	0
Name	swapper / sched
Created by	Kernel (at boot)
Runs in	Kernel space (not user space)
Visible?	Usually nahi (but <code>ps -ef</code> ya <code>top</code> me kabhi kabhi dikhta hai)

Bonus Info:

- PID 1 hota hai init ya systemd, jo user space ka first process hota hai.
- PID 0 kernel ke andar hota hai, aur koi normal process nahi hota jese baaki.

top kya hai

top command kya karta hai?

- top आपको batata hai ki kaunse processes system resources (CPU, RAM, etc.) sabse zyada use kar rahe hain — bilkul live monitoring jaisa.

top command ka output mein key cheezein:

- 1. System summary line (top section):

```
top - 14:02:53 up 1:15, 2 users, load average: 0.05, 0.12, 0.09
```

- Time, Uptime, Users, Load average (1, 5, 15 min)

2. Tasks:

```
Tasks: 152 total, 1 running, 151 sleeping, 0 stopped, 0 zombie
```

- Kitne total processes hain, kaunse running/sleeping hain

3. CPU usage:

```
%Cpu(s): 3.2 us, 0.5 sy, 0.0 ni, 96.1 id, 0.1 wa, 0.0 hi, 0.1 si, 0.0 st
```

- us: user space usage
- sy: system (kernel) usage
- id: idle
- wa: wait for I/O

4. Memory and Swap:

MiB Mem : 7880.6 total, 2500.0 free, 3100.5 used, 2280.1 buff/cache MiB Swap: 2048.0 total, 1024.0 free, 1024.0 used

5. Processes list (bottom section):

```
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND 1234 root 20 0 500000 10000 8000 S
5.3 0.2 0:01.23 firefox
```

- PID: Process ID
- USER: Process ka owner
- %CPU: Kitna CPU use ho raha hai
- %MEM: Kitna memory use ho raha hai
- COMMAND: Process ka naam

Tip:

- Agar aapko better visual chahiye to htop use kar sakte hain (install separately):

```
apt install htop
htop
```

Process ke Types:

Type	Description
Foreground	Jo aapke terminal me directly run hota hai
Background	Jo terminal se detach ho kar silently run karta hai

note:

hhttpd

- is ko apachi khte ha

mysql -is ko data base khte hai docker

- is ko container server khte ha

Open file

- jo be hmary file ya parogram loaded huta hai RAM me to is ko open file khte hai

Isof

- is ki madat se hum dekh skte ha hmary pass ktne open file hai

grep kya karta hai?

- grep kisi file ya command ke output me se specific pattern ya keyword dhoondhta hai aur sirf matching lines print karta hai.

Syntax:

```
grep [options] "search-pattern" filename
```

Examples:

- 1. File me se ek word dhoondhna:

```
grep "hello" file.txt
```

- file.txt me jo bhi lines "hello" contain karti hain, wo print hongy.
- 5. Command ke output me se filter:

```
ps aux | grep ssh
```

- ps output me se sirf ssh related processes dikhayega.

color k sath find karna hu

```
grep --color "find word name" /etc/passwd
```

pipe se grep

```
ls /etc/* | grep -i host
```

(|) ya kia hai pipe

Pipe (|) kya hai?

Definition:

- (|) ka use ek command ka output ko dusri command ke input ke roop mein dene ke liye hota hai.

- ska matlab: "Jo pehle command ka output hai, use agle command ka input bana do."

Basic syntax

```
command1 | command2
```

Examples:

- Example 1: Is output ko grep ke saath filter karna

```
ls -l | grep ".txt"
```

Pipe ka Real Use:

- Command chaining
- Filters and transformations
- Logs parsing
- Data stream handling

Extra Tip:

- Pipe sirf standard output (stdout) ko standard input (stdin) mein pass karta hai. Agar aap error (stderr) bhi pipe karna chahte hain, to aapko `2>&1` jaisa redirection use karna hota hai.

Example:

```
command1 2>&1 | command2
```

```
ls /etc/* | less
```

- tmam data ak page me data hai

```
ls /etc/* | more
```

- ye command page wise scrole karta hai

```
ls | wc -l
```

- ye words count karte hai

find and locate commands

1. find Command

Definition:

- find command aapko real-time file system traversal karke files/directories search karne deta hai.

```
find [path] [options] [expression]
```

Examples:

Search by file name:

```
find /home -name "file.txt"
```

- Search by permission:

```
find / -perm 777
```

Pros:

- Real-time accurate results
- Flexible (name, size, time, type, etc.)
- Can run actions like -delete, -exec, etc.

2. locate Command

Definition:

- locate command ek pre-built file database ko use karta hai jo system mein already indexed hoti hai (usually /var/lib/mlocate/mlocate.db).

Syntax:

```
locate [filename]
```

Note:

- locate ka database regularly update hota hai updatedb command ke through. Agar file nayi ho aur database update na ho, to locate use nahi dhoondh payega.

Difference between find and locate:

Feature	find	locate
Speed	Slow (real-time search)	Fast (pre-indexed database)
Accuracy	Highly accurate (live file system)	May miss new files
Flexibility	High (many search options)	Limited (name/path based)
Database use	No	Yes (mlocate.db)
Requires sudo	Often yes	Rarely
Real-time data	Yes	No

- locate database update (manual):

updatedb

Summary:

Command	Use
<code>find /home -name "file.txt"</code>	Real-time file search
<code>locate file.txt</code>	Fast search from indexed database